MUHAMMAD

Lead Data Architect & Senior Data Science and Data Engineer

Lahore, Pakistan • Open to Remote Engagements

SUMMARY

Senior Data Engineer and Lead Data Architect with 8+ years of expertise designing, building, and optimizing enterprise-grade cloud-native data platforms, high-volume ETL/ELT pipelines, and AI-ready data infrastructure. Top-rated Upwork freelancer with $400K+ earned across 50+ global engagements. Deep hands-on experience across AWS (S3, Glue, Redshift, Lambda, EMR, DMS, SageMaker, Bedrock), Azure (ADF, Synapse, Databricks, ADLS), Snowflake, PySpark, Databricks, Delta Lake, Apache Airflow, dbt, and GitHub-based CI/CD pipelines. Applied machine learning across multiple client engagements — clustering, logistic regression, random forests, SVM, and neural networks — spanning record classification and entity resolution, behavioural segmentation, forecasting, and computer-vision image classification. Proven track record leading cross-functional engineering teams, architecting lakehouse and data warehouse solutions, and delivering measurable business impact — including $4M+ in annual savings, 40% DWH performance gains, and 50% reduction in manual pipeline operations. Recognized for production-grade reliability standards, data governance, and the ability to translate complex data requirements into scalable, maintainable systems.

TECHNICAL SKILLS

AWS	S3, Glue, Glue Data Catalog, Redshift, Athena, EMR, Lambda, DMS, RDS, Lake Formation, Kinesis, Step Functions, SageMaker, Bedrock, Kendra, Textract, Comprehend, OpenSearch, CloudWatch, IAM, KMS, CloudFormation, S3 Glacier, EBS
Azure	Azure Data Factory (ADF), Azure Synapse Analytics, Azure Data Lake Storage (ADLS), Azure Databricks
Machine Learning	Clustering (K-Means, hierarchical, DBSCAN), Logistic Regression, Random Forests, Support Vector Machines (SVM), Neural Networks (CNNs for image classification), feature engineering, model training & evaluation, segmentation, classification, forecasting
Data Warehousing	Snowflake, Amazon Redshift, Google BigQuery, SQL Server, Delta Lake (ACID compliance, schema enforcement)
Data Engineering	PySpark, Apache Spark, Spark SQL, DataFrame APIs, Apache Airflow (DAG design, SLA monitoring), dbt (incremental models, materializations), Fivetran, Talend, Apache NiFi, Apache Kafka, Apache Hive, Apache Pig
Lakehouse & Streaming	Databricks, Delta Lake, Apache Kafka, Amazon Kinesis, Amazon MSK, Parquet, ORC, JSON
CI/CD & DevOps	GitHub, GitLab, Jenkins, Git-based deployment pipelines, Docker, Kubernetes, Ansible, CloudFormation, Infrastructure-as-Code
Orchestration	Apache Airflow, Rundeck, AWS Step Functions, Apache NiFi — dependency management, retries, SLA alerting
BI & Analytics	Tableau, Power BI, Amazon QuickSight, Looker, Apache Druid, Plotly, Dash, Pandas
Databases & Search	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Amazon OpenSearch, Amazon DynamoDB
Programming	Python, SQL, PL/SQL, Batch Scripting, Node.js, Java, C#, Flask, Django, SQLAlchemy
Data Quality & Governance	SLA/SLO tracking, data quality validation frameworks, pipeline health checks, audit logging, RBAC, Lake Formation permissions, KMS encryption, S3 bucket policies
AI / ML Data	SageMaker Feature Store, Bedrock Knowledge Bases, RAG pipelines, NLP, document intelligence, Kendra, Textract, Comprehend
Marketing Data	Tealium, attribution modeling, campaign analytics, denormalized data marts, conversion prediction

EXPERIENCE

Lead Data Architect | Dot Labs FZC

Sharjah, UAE (Remote) • 2023 – Present

- Lead Data Architect across 50+ global client engagements, owning end-to-end architecture, delivery, and quality across AWS, Azure, and GCP data platforms.
- **Aquascope (AWS · AppSync · Claude · MCP):** Architected and operated the complete backend and data engineering infrastructure on AWS; integrated AppSync for real-time GraphQL APIs; implemented MCP-based tooling with Claude to deliver intelligent, AI-driven product features on a production-ready, AI-ready backend.
- **COVID-19 Data Classification & Kinship Identification (Clustering · Classification · Python):** Built machine learning models over large-scale COVID-19 datasets to classify records and identify kinship/household relationships between individuals; combined unsupervised clustering with supervised classifiers (Logistic Regression, Random Forests, SVM) for entity resolution, engineered demographic and locality features, and validated results through precision/recall and cluster-quality evaluation.
- **Home Image Classification (Neural Networks · Computer Vision):** Developed convolutional neural network models to automatically classify and tag property/home images by room type and condition, replacing manual tagging workflows and enabling searchable, structured image metadata at scale.
- **Traffic Analysis & Prediction (Machine Learning · Python):** Delivered traffic pattern analysis and prediction models over high-volume sensor and telemetry data — applied clustering for pattern discovery and regression/ensemble models (Logistic Regression, Random Forests, SVM) for classification and forecasting, surfacing actionable congestion and flow insights.
- **nOps (Apache Druid · Performance Optimization):** Optimized Apache Druid environment for high-volume analytical workloads — tuned ingestion strategies, segment design, and cluster configuration to significantly reduce query latency and cost, enabling faster real-time analytics at scale.
- **Sindri Media (Airflow · dbt · RDS):** Designed ELT pipelines orchestrated via Apache Airflow with dbt-driven transformations on RDS; standardized data models for analytics-readiness and established a maintainable, well-documented data stack.
- **Retailogist (Data Engineering · European Trading):** Built data engineering and analytics pipelines to consolidate and structure trading data for a European commodities company; improved data visibility and reporting accuracy across operations.
- **Lemmata (Google Cloud Platform):** Designed and implemented cloud-native data engineering pipelines on GCP, building scalable workflows tailored to client data and analytics requirements.
- **Harwell (Azure · ADF · Synapse):** Delivered end-to-end Azure data engineering using Azure Data Factory for orchestration and Synapse Analytics for warehousing; established a robust analytics foundation on the Microsoft cloud stack.
- **Hospi-Housing (AWS · QuickSight):** Built analytics and reporting infrastructure on AWS, delivering interactive Amazon QuickSight dashboards turning raw operational data into self-serve business insights.
- **Woodford (Power BI · Salesforce):** Integrated Salesforce data into Power BI, building executive dashboards for unified CRM and operational performance visibility.
- **Custom-DB – Commodities Trading (Airflow · Python):** Automated ingestion, cleaning, and delivery of cotton and coffee trade datasets using Apache Airflow and Python to power reliable market intelligence pipelines.
- Managed CI/CD pipelines using GitHub, GitLab, and Jenkins across client engagements; enforced Git-based deployment workflows, Docker containerization, and infrastructure-as-code with CloudFormation and Ansible.
- Led engineering team overseeing delivery quality, architecture reviews, code standards, and client success across concurrent engagements.

Data Engineer | Upwork
USA (Remote) • 2022 – 2023

- Worked within Upwork's internal data engineering team on a modern cloud stack — Snowflake, AWS, Apache Airflow, and Rundeck — building and orchestrating scalable data pipelines supporting high-volume platform analytics.
- Integrated Tealium for marketing data ingestion and ads analytics; designed attribution models and data mart schemas that increased marketing ROI by 20%.
- Drove $4M+ in annual savings by enhancing campaign attribution and conversion prediction models using ML-powered segmentation — applied clustering and logistic regression to identify high-intent user cohorts and predict conversion likelihood.
- Built denormalized data structures in Snowflake for real-time marketing campaign support; established data quality checks across 100+ ingestion connectors ensuring warehouse integrity.
- Managed 50+ data pipelines using Rundeck and Airflow; automated reporting and forecasting workflows, reducing manual data processing by 50%.
- Implemented and maintained CI/CD pipelines using GitHub Actions and GitLab for data pipeline deployments; enforced version control and Git-based workflows across the engineering team.
- Collaborated with product and data science teams to deliver ML-powered keyword clustering and user segmentation tools.

Data Engineer — Flywheel Project | Strategic Systems International
Lahore, Pakistan • 2021 – 2022

- Managed and engineered data infrastructure handling 1+ Petabyte of data for a unified retail analytics platform consolidating brand data from Amazon, Shopify, Instacart, and Walmart into a single reporting layer.
- Designed and maintained production ETL pipelines using AWS Glue, Apache Airflow, and Lambda; used Amazon Redshift as the primary data warehouse and AWS DMS for large-scale data migration tasks.
- Implemented CloudFormation-based infrastructure-as-code and GitHub CI/CD workflows for pipeline deployments; enforced environment consistency and reduced configuration drift across production systems.
- Enforced data quality validation, SLA tracking, and incident-response workflows to maintain production reliability across petabyte-scale pipelines.
- Core stack: AWS Redshift, AWS Glue, AWS Lambda, AWS RDS MySQL, AWS DMS, Apache Airflow, Python.

Software Engineer – Data | Kavtech Solutions (Private) Ltd.
Lahore, Pakistan • Feb 2020 – Oct 2021

- Led a team of 8 engineers on the Ylopo real estate client engagement; optimized data warehouse performance by 40% and increased team productivity by automating manual data workflows.
- Designed DWH architecture and optimized queries to extract data from multiple OLTP systems; built real estate data integration pipelines consuming structured and unstructured API data for agent-facing platforms.
- Established Git-based version control and deployment practices across the team; introduced code review standards and CI workflows to improve delivery quality.
- Technologies: PostgreSQL, Pentaho, Tableau, AWS Athena, S3, Java.

Software Engineer (Part-time) | Kavtech Solutions (Private) Ltd.
Lahore, Pakistan • Jun 2017 – Jan 2020

- Developed a desktop-based HRMS covering employee management, payroll, RFID attendance, leave, project tracking, and interview management; delivered in ASP.NET, C#, and MS SQL using Agile methodology.
- Automated HR reporting workflows including employee letters, clearance documents, attendance reports, and performance reviews using Crystal Reports.

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services

EDUCATION

Bachelor's Degree, Computer Science | FAST National University of Computer & Emerging Sciences
Lahore, Pakistan • 2016 – 2020

SOFT SKILLS

Effective Communication • Team Leadership • Client-Centric Approach • Analytical Thinking • Empathy & Emotional Intelligence • Continuous Learning • Creativity & Innovation